Q436 and Q435: the branch assignment, and what $t = 0$ actually counts

Warnaar's t grades the winding number, not a Hall–Littlewood parameter

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Abstract

The session was briefed as a read plus arithmetic: assign the Macdonald branch of the target $c_{(1^4)}(t) = 3 - t^2$ (Q436), then test the hypothesis that $c_\alpha(0)$ is an undeformed coset count (Q435). Both questions are answered, and both answers correct the brief. (i) Korff–Palazzo `1804.05647` does *not* determine the branch: it names the q-Whittaker specialisation once, without a display, and its own letter t is a third parameter. The primary source Korff `1110.6356` §2.6 gives both specialisations explicitly, and his $f^{\nu}_{\lambda\mu}(t)$ is Hall–Littlewood. (ii) The branch assignment is nevertheless *irrelevant to every weight kill*, because both specialisations have multiplicity-determined single-box skew weights and are therefore both killed by Theorem D of 2026-10-12 (Theorem 10 below, 280/280). (iii) The real finding is Theorem 13: in Warnaar's $\mathrm{Eq}_{\mathrm{generalise}}$ every t-exponent is non-negative and vanishes only at zero winding, so $t = 0$ selects the *unwrapped* determinant. Hence $c_\alpha(0)$ is the count of zero-winding tableaux (Theorem 14, 1656/1656), the undeformed *cylindric* count is $c_\alpha(1)$, and the briefed hypothesis is refuted on its pre-registered arm A2 with the discrepancy located at the normalisation end. Warnaar's t sits in Korff's winding slot z, not in his Hall–Littlewood slot t nor in his q-Whittaker slot q — which is a structural explanation of why four successive per-box weight architectures died.

§0 Holdings, measured by this session

Root list searched, printed because a holdings verdict's domain *is* its root list: `/home/clio/data`, `/home/clio/projects`, `/home/clio/git`, `/home/clio/mail`. Every file below was `stat`-ed by this session at 13 October, start of slot; sizes are bytes as measured, not copied from the brief.

paper	path measured present	byte
Korff–Palazzo `1804.05647`	`projects/sources/20260924c1-novelty/korff-palazzo-1804.05647.tex`	235,42
Morse–Schilling `1207.0786`	`projects/library/.../wk/ms-1207.0786.pdf`	109,84
Korff `1110.6356` (pdf)	`projects/library/.../wk/korff-1110.6356.pdf`	1,126,69
Korff `1110.6356` (text layer)	`projects/library/.../wk/korff-1110.6356.txt`	202,04
HKKO `2301.13117`	`/home/clio/data/fetched/arxiv-2301.13117.pdf`	662,21
Wheeler–Zinn-Justin `1603.01815`	`git/puzzles/seed-papers/wheeler-zinn-justin-2016-...pdf`	665,58

All six present. Line numbers quoted below refer to the `1110.6356` text layer (`korff-1110.6356.txt`, 7,650 lines, 38,506 words) and to the `1804.05647` LaTeX source (2,853 lines), both read first-hand this session. Morse–Schilling was re-extracted this session with `pdftotext` (4,740 words, 700 lines).

1 §1 Q436: the branch assignment

1.1 What Korff–Palazzo `1804.05647` does and does not say

Measurement 1. `1804.05647` is Korff–Palazzo, *Cylindric symmetric functions and positivity* (title, l. 132). The string `Whittaker` occurs **exactly once** in the whole 25,851-word source (`grep -ci whittaker` = 1), at l. 182, in the introduction, describing a *different* paper:

> "... a q-deformation of the $\widehat{sl}_n$-Verlinde algebra was constructed in [korff2013cylindric] which ... whose structure constants or fusion coefficients $N^{\nu}_{\lambda\mu}(q) \in \mathbb{Z}[q]$ are related to cylindric versions of Hall–Littlewood and q-Whittaker polynomials. Both types of polynomials occur (in the non-cylindric case) as specialisation of Macdonald polynomials." (l. 182)

No display, no substitution into $P_\lambda(x;q,t)$. The cited `korff2013cylindric` is, from the bibliography (ll. 2705–2709), Korff, *Cylindric versions of specialised Macdonald functions and a deformed Verlinde algebra*, CMP **318** (2013) 173–246 = arXiv `1110.6356`.

Measurement 2 (answering §1(iv))**.** *Palazzo's t is a third parameter.* In `1804.05647` the letter t is a cyclotomic Hecke/twist variable, not a Macdonald parameter: l. 429 defines the deformed $\mathcal{S}$-matrix $\mathcal{S}(t) = \frac{1}{\sqrt{n}} \sum_{i,j} t^{-j} \zeta^{ij} e_{ij}$ with $\zeta = \exp(2\pi\iota/n)$ (l. 334); the lemma after it sets $z = t^n$; and l. 684 is the cyclotomic polynomial $f(x_1) = (x_1 - t)(x_1 - t\zeta)\cdots(x_1 - t\zeta^{n-1}) = x_1^n - t^n$. At l. 953 this t multiplies the *undeformed* fusion coefficient as a pure grading factor, $\langle v^\nu, M^*_\lambda v_\mu \rangle = t^{|\nu|-|\lambda|-|\mu|} N^\nu_{\lambda\mu}$, with $N^\nu_{\lambda\mu} = 0$ unless $|\lambda| + |\mu| - |\nu| \equiv 0 \bmod n$ (l. 955). The body of the paper works at $q \to 1$ (l. 184: "we investigate the combinatorial structure of the q-deformed Verlinde algebra from [korff2013cylindric] **in the limit** $q \to 1$").

Corollary 3 (§1(i), answered in the negative)**.** *Korff–Palazzo `1804.05647` does not determine the branch of any t-dependent target. It mentions the two specialisations once without displaying either, works only at $q \to 1$, and uses the letter t for an unrelated cyclotomic parameter.* This is a result, not a failure of the read — *the brief named the wrong paper, and the right one is on disk.*

1.2 The two specialisations, from the primary source

Measurement 4 (§1(i) from `1110.6356` §2.6, read first-hand)**.** Both specialisations are displayed in Korff `1110.6356` §2.6.

- **Hall–Littlewood branch: $q = 0$, surviving parameter t.** "In what follows we will omit the dependence on the second parameter in the notation and simply write $Q_\lambda(x;t) = Q_\lambda(x;0,t)$ and $P_\lambda(x;t) = P_\lambda(x;0,t)$" (l. 1313). The skew weights are, eq. (2.21),

$$\psi_{\lambda/\mu}(t) = \prod_{i \in J_{\lambda/\mu}} \left(1 - t^{\mu'_i - \mu'_{i+1}}\right), \qquad J_{\lambda/\mu} = \{\, i : \theta'_i = 0,\ \theta'_{i+1} = 1 \,\}, \quad \theta' = \lambda' - \mu',$$

  zero unless λ/μ is a horizontal strip, with $m_i(\mu) = \mu'_i - \mu'_{i+1}$ the multiplicity of the part i (l. 1296).

- **q-Whittaker branch: $t = 0$, surviving parameter q.** "In light of the definition (2.19) it is natural to consider also the complementary limit of Macdonald polynomials setting now $t = 0$" (l. 1349). Eq. (2.24):

$$\psi'_{\lambda/\mu}(q) := \psi_{\lambda/\mu}(q,0) = \prod_{i>0} \begin{bmatrix} \lambda_i - \lambda_{i+1} \\ \lambda_i - \mu_i \end{bmatrix}_q ,$$

again zero off horizontal strips, with the companion $\varphi'_{\lambda/\mu}(q)$ at eq. (2.25). Korff adds (l. 1427): "In [26] they have been named q-deformed or simply q-Whittaker functions."

Measurement 5 (§1(ii)). *Korff's $f^{\nu}_{\lambda\mu}(t)$ is on the Hall–Littlewood branch.* It is introduced in §2.6.2, titled "Hall algebra, coproduct and skew Hall–Littlewood functions", as the coefficients in the product expansion (1.3) of Hall–Littlewood P-functions, and Example 7.2 calls them "the expansion coefficients in the corresponding product of Hall–Littlewood polynomials" (l. 7230). The q-Whittaker objects carry primes (ψ', φ', P', Q') and the letter q throughout. There is no ambiguity.

Measurement 6 (the brief's control, corrected). **f and N are different objects, and the brief's "undeformed fusion/coset count" is a Littlewood–Richardson coefficient.** In Example 7.2 ($n = 4$, $k = 3$, $z = 1$, $\lambda = (3,2,1)$, $\mu = (4,3,1)$) Korff writes, l. 7177:

> "We exploit the fact that $f^{\nu}_{\lambda\mu}(t) \equiv 0$ unless $f^{\nu}_{\lambda\mu}(0) = c^{\nu}_{\lambda\mu} \neq 0$ **where $c^{\nu}_{\lambda\mu}$ is the Littlewood–Richardson coefficient**."

So the briefed "$c^{(6,5,3)}_{\lambda\mu} = 2$" is an LR coefficient. The *deformed fusion* coefficients are the separate family $N^{\lambda}_{\mu\nu}(t)$ obtained from the f's by signed straightening (ll. 7256–7262), and for them Korff states, l. 7266: "Setting $t = 0$ one can verify, using the Verlinde formula or the Kac–Walton algorithm, that one obtains the correct fusion coefficients of the $\widehat{sl(n)}_k$-Verlinde algebra."

1.3 Firing control: the pipeline reproduces Korff's printed values

The brief required that any pipeline be run first on a case whose answer is printed. All inputs below are quoted from the text layer; nothing is reconstructed from Korff's definitions.

ν	$c^{\nu}_{\lambda\mu}$ (LR, l. 7204)	$f^{\nu}_{\lambda\mu}(t)$ (l. 7236)	straightening (l. 7244)
$(7,5,2)$	1	1	$P_{(7,5,2)} = t\,P_{(3,2,1)}$
$(7,4,3)$	1	1	$P_{(7,4,3)} = [2]\,P_{(4,3,3)}$
$(6,6,2)$	1	$1+t$	$P_{(6,6,2)} = [3]\,P_{(2,2,2)}$
$(6,5,3)$	2	$2-t^2$	$P_{(6,5,3)} = P_{(3,2,1)}$
$(6,4,4)$	1	$1+t$	$P_{(6,4,4)} = P_{(4,4,2)}$
$(5,5,4)$	1	$1+t$	$P_{(5,5,4)} = P_{(4,1,1)}$

Proposition 7 (Control 1, 6/6). *$f^{\nu}_{\lambda\mu}(0) = c^{\nu}_{\lambda\mu}$ for all six ν. This is not a coincidence to be checked but a consequence of $P_{\lambda}(x;0,0) = s_{\lambda}$; Korff states it at l. 7177.*

Proposition 8 (Control 2, 5/5: the pipeline fires). *Collecting the straightened terms reproduces every printed N:*

$$N^{(3,2,1)} = t\cdot f^{(7,5,2)} + 1\cdot f^{(6,5,3)} = t + (2-t^2) = 2 + t - t^2, \quad N^{(2,2,2)} = [3]\,f^{(6,6,2)} = (1+t+t^2)(1+t),$$
$$N^{(4,3,3)} = [2]\,f^{(7,4,3)} = 1+t, \quad N^{(4,4,2)} = f^{(6,4,4)} = 1+t, \qquad N^{(4,1,1)} = f^{(5,5,4)} = 1+t,$$

agreeing with Korff's printed list in all five entries. A consistency check on the same display: "after removing all n-rows" ($n = 4$) his expansion reads $(2+t-t^2)P_{(3,2,1)} + (1+t)(1+t+t^2)P_{(2,2,2)} + (1+t)\big(P_{(3,3,0)} + P_{(1,1,0)} + P_{(2,0,0)}\big)$, and the three last terms are exactly $(4,3,3) \to (3,3,0)$, $(4,1,1) \to (1,1,0)$, $(4,4,2) \to (2,0,0)$ with the common coefficient $1+t$. Everything closes.

Remark 9 (the brief's second control, honestly graded). The brief proposed $N^{(3,2,1)} = 2+t-t^2$ as a case "where $f(0) = 2$ but the polynomial has a linear term, so it distinguishes 'H about the constant term' from 'H about the whole polynomial'". It does not: $N^{(3,2,1)}(0) = 2$ and $N^{(3,2,1)}(1) = 2$ as well, so the two readings agree on it. What the case *does* separate is f from N: they share the constant term 2 and differ by exactly $t\cdot f^{(7,5,2)}$, the straightening of $P_{(7,5,2)} = t\,P_{(3,2,1)}$. That is a sharper statement than the one asked for, and it is the one Proposition 8 establishes.

1.4 §1(iii): the branch assignment, and why it changes nothing

The brief's worry was that if the target sits on the q-Whittaker branch then "some of the four kills were killing the right mechanism on the wrong target". That worry is unfounded, for a reason sharper than a branch assignment.

Recall Theorem D (registry node `multiplicity-determined-architecture-refuted`, `proved`, 2026-10-12): for $k \geq 1$, $\ell = 0$ (so $h = 2k$, $w = 2$), $\alpha = (1, 1)$, the fibre has exactly two cylindric tableaux, with $c^- = +1$ and $c^- = -1$, and with *identical* multiplicity-value sequences $(2, 1)$, differing only in the cyclic *index* at which each value is read ($(h, 1)$ versus (h, h)). Hence for every weight of the form $W_T = \prod f(m)$ — a *multiplicity-determined* multiplicative weight — $\sum_T c^-(T) W_T = 0$ identically, while $c_{(1,1)}(t) = 1 - t \neq 0$.

Theorem 10 (both Macdonald branches are multiplicity-determined on a single box)**.** *Let λ/μ be a single box.*

1. (Hall–Littlewood.) $|J_{\lambda/\mu}| \leq 1$, *and* $\psi_{\lambda/\mu}(t) = 1 - t^m$ *with* $m = \mu'_{c-1} - \mu'_c$ *a single multiplicity, where c is the column of the box (and $\psi = 1$ when $c = 1$).*
2. (q-Whittaker.) $\psi'_{\lambda/\mu}(q) = \left[\lambda_j - \lambda_{j+1}\right]_q = 1 + q + \cdots + q^{m-1}$ *with* $m = \lambda_j - \lambda_{j+1}$ *a single gap, where j is the row of the box.*

In both cases the weight is $f(m)$ for a function f independent of the index. *Hence both branches are multiplicity-determined, and Theorem D refutes both.*

Proof. (1) If the box lies in column c then $\theta' = \lambda' - \mu' = e_c$, so $\theta'_i = \delta_{i,c}$ and the condition $\theta'_i = 0,\ \theta'_{i+1} = 1$ holds exactly for $i = c - 1$. Thus $J_{\lambda/\mu} = \{c - 1\}$ if $c \geq 2$ and $J_{\lambda/\mu} = \varnothing$ if $c = 1$, giving a single factor $1 - t^{\mu'_{c-1} - \mu'_c}$ (resp. the empty product).

(2) If the box lies in row j then $\lambda_i - \mu_i = \delta_{ij}$. The Gaussian binomial $\binom{a}{0}_q = 1$ kills every factor $i \neq j$, and the factor $i = j$ is $\binom{\lambda_j - \lambda_{j+1}}{1}_q = \frac{1 - q^{\lambda_j - \lambda_{j+1}}}{1 - q} = [\lambda_j - \lambda_{j+1}]_q$.

In (1) the function is $f(m) = 1 - t^m$, in (2) it is $f(m) = [m]_q$; neither depends on c resp. j except through the value m. Theorem D applies verbatim. □

Verification (symbolic). All 280 single-box pairs $(\mu, \mu + e_j)$ with $\ell(\mu) \leq 4$ and parts ≤ 4: the Hall–Littlewood claim holds 280/280, with 224 of the 280 having $|J| = 1$ (so the non-trivial case is exercised, not vacuous); the q-Whittaker claim holds 280/280, with 140 having $m \geq 2$ (so the non-constant case is exercised). Evaluating the four candidate weights on the two-tableau fibre:

Korff 2011 $\psi : f(m) = 1 - t^m$:	$W_{T_1} = W_{T_2} = (1 - t^2)(1 - t)$,	$\sum c^- W_T = 0$,
Korff 2019 ribbon $wt : f(m) = 1$:	$W_{T_1} = W_{T_2} = 1$,	$\sum c^- W_T = 0$,
q-Whittaker $\psi' : f(m) = [m]_q$:	$W_{T_1} = W_{T_2} = 1 + q$,	$\sum c^- W_T = 0$,
Q430 Gaussian $f(m) = [m + 1]_t$:	$W_{T_1} = W_{T_2} = [3]_t [2]_t$,	$\sum c^- W_T = 0$,

against the target $c_{(1,1)}(t) = 1 - t \neq 0$ on that fibre: four kills, one evaluation each.

Satisfiability control (mandatory, per the Q430 rule). A class-kill is indistinguishable from a broken fibre unless something in the widened class succeeds. An index-aware weight does: taking $g(m{=}2, i{=}h) = 1$, $g(m{=}1, i{=}1) = 1$, $g(m{=}1, i{=}h) = t$ gives $\sum_T c^-(T) W_T = 1 - t$, hitting the target exactly. The fibre is not broken; the architecture is.

Remark 11 (honest grading of the verification)**.** The four-row evaluation above is an *evaluation*, not an independent verification: the two W_T are equal *by construction*, because Theorem D's two-tableau structure (registry node `two-tableau-fibre-structure`, `proved`, verified symbolically for $k = 1, \dots, 7$) already says the m-value sequences coincide. The independent content is Theorem 10: that ψ' — not previously tested — lands in that class. The 280/280 sweep is the check that bears weight.

Corollary 12 (§1(iii), answered)**.** *The branch assignment of $c_{(1^n)}(t)$ cannot rescue any of the four weight-architecture kills* (Q405, Q410, Q417, Q430)*, because the per-box skew weights of both branches are multiplicity-determined and both die on a two-tableau fibre. Moreover the four kills are arithmetic statements about a one-variable polynomial, and each branch carries exactly one parameter, so relabelling $t \leftrightarrow q$ changes none of them.*

2 §2 Q435: the pre-registered arms, and what $t = 0$ counts

2.1 The arms, as written in the brief, before their outcomes

The brief pre-registered, on hypothesis **H:** $f^{\nu}_{\lambda\mu}(t = 0)$ *equals the undeformed fusion/coset count* $c^{\nu}_{\lambda\mu}$, transported to the target so that "the 3 is a coset count and the $-t^2$ is entirely the deformation":

A1 (H confirmed) the alternating sum at the point corresponding to $k = 2$, $\ell = 1$, $\alpha = (1^4)$ evaluates to **3** $\Rightarrow$ branch confirmed, $-t^2$ isolated as pure deformation, problem halved.

A2 (H refuted) it evaluates to anything $\neq 3$ $\Rightarrow$ the dream's sentence is false; record which value, and whether the discrepancy is the ℓ-shift, the level–rank dual, or the dictionary. **A2 is the more informative arm.**

A3 (undecidable from the abstract) the 12-pp extended abstract omits the construction $\Rightarrow$ stop, record a reading debt, do not reconstruct.

Outcome: A2, and A3 does not fire. The value is **2**, not 3; the discrepancy is *none of the three listed causes*; and it is the normalisation end.

2.2 The theorem: t grades the winding number

Warnaar's generalised determinant (`2511.17034`, $\mathrm{Eq}_{\mathrm{generalise}}$), in the normalisation of the Q405 engine with $M = k + \ell + 1$, $N = 2M = 2k + w$, $w = 2\ell + 2$:

$$G_t = \sum_{y \in \mathbb{Z}^k} \det_{1 \le i,j \le k} \Big(t^{My_i^2 - jy_i} \big(\dot{e}_{n-i+j-Ny_i}(x) - \dot{e}_{n+i+j-Ny_i}(x) \big) \Big), \qquad \dot{e}_r(x) = e_r\big(x_1, x_1^{-1}, \dots, x_n, x_n^{-1}\big), \tag{1}$$

and $c_\alpha(t) := [x^\alpha]\,(x_1 \cdots x_n)^k G_t$. The vector y is the vector of winding numbers around the cylinder.

Theorem 13 ($t = 0$ selects zero winding)**.** *Every t-exponent occurring in (1) is a non-negative integer, and it vanishes exactly on the summand $y = 0$. Consequently*

$$G_0 = \det_{1 \le i,j \le k} \big(\dot{e}_{n-i+j}(x) - \dot{e}_{n+i+j}(x) \big),$$

the single $y = 0$ term: the unwrapped *(non-affine) bounded Littlewood determinant. Equivalently, in HKKO's form, $t = 0$ replaces each affine periodisation $F_{r,N} = \sum_{m \in \mathbb{Z}} f_{r+Nm}$ by its single term f_r.*

Proof. Expanding the determinant over permutations $\sigma \in S_k$, a term carries the exponent $\sum_{i=1}^{k}\big(My_i^2 - \sigma(i)y_i\big)$, so it suffices to show each summand $\varepsilon(y,j) := My^2 - jy = y(My-j)$ is ≥ 0 for $1 \leq j \leq k$, with equality iff $y = 0$.

If $y = 0$ then $\varepsilon = 0$. If $y \geq 1$ then $My - j \geq M - k = \ell + 1 \geq 1 > 0$, so $\varepsilon = y(My-j) \geq 1 > 0$. If $y \leq -1$ then $y^2 \geq 1$ and $-jy \geq j \geq 1$, so $\varepsilon = My^2 - jy \geq M + 1 > 0$. Hence every exponent is ≥ 0, and the total exponent vanishes iff $y_i = 0$ for all i. Setting $t = 0$ therefore annihilates every summand with $y \neq 0$ and leaves the $y = 0$ determinant, whose entries are $\dot{e}_{n-i+j} - \dot{e}_{n+i+j}$.

For the last sentence: with $\dot{e}_{n-r} = (x_1 \cdots x_n)^{-1} f_r$ and $f_r = f_{-r}$, one has $(x_1 \cdots x_n)^k \det_k(\dot{e}_{n-i+j} - \dot{e}_{n+i+j}) = \det_k(f_{i-j} - f_{i+j})$, which is HKKO's $\det_k(F_{-i+j,N} - F_{i+j,N})$ with each F truncated to its $m = 0$ term. □

Verification. $G(t{=}0)$ computed with t substituted *before* forming the determinant, and G_t expanded symbolically then evaluated at $t = 0$, both compared against the $y = 0$ determinant: agreement on all 10 triples $(k,\ell,n) \in \{(1,0,2),(1,0,4),(1,0,5),(1,1,4),(1,1,5),(1,2,4),(2,0,2),(2,0,4),(2,1,3),(2,1,4$ i.e. 20/20 comparisons. The inequality $\ell + 1 \geq 1$ in the proof is why $M > k$ matters: the hypothesis $1 \leq j \leq k < M$ is exactly Warnaar's $M = k + \ell + 1$ with $\ell \geq 0$.

2.3 What $c_\alpha(0)$ counts

Write A_α, B_α, TOT_α for the numbers of cylindric tableaux of content α — lattice paths in $\mathbb{Z}^{2k}$ from 0 with step i a 0/1 vector of weight α_i, staying in $\{x_1 \geq \cdots \geq x_{2k} \geq x_1 - w\}$ — whose endpoint λ has $c^-_{2k,w}(\lambda) = +1$, -1, and all of them. Let A^∞_α be the same count with the *width bound* $x_{2k} \geq x_1 - w$ dropped from the path condition (the sign rule c^- itself unchanged).

Theorem 14 (what the two evaluation points count)**.**

$$c_\alpha(1) = A_\alpha - B_\alpha \quad \textit{(HKKO, } \texttt{t1-gate}\textit{, } \texttt{proved}\textit{)}, \qquad\qquad c_\alpha(0) = A^\infty_\alpha.$$

In particular $c_\alpha(0) = A_\alpha$ iff the width bound is inactive on the fibre*, and* $c_\alpha(0)$ *is in general* ***not*** *a count of objects in the cylindric fibre at all: it can exceed* TOT_α*.*

Proof of the second identity, modulo the stated gap. By Theorem 13, $(x_1 \cdots x_n)^k G_0 = \det_k(f_{i-j} - f_{i+j})$, which is HKKO's left-hand side with the affine periodisation removed. HKKO's identity (`2301.13117`, Thm 3.3) equates the periodised determinant with $\sum_{\lambda \in \mathrm{Par}(2k,w)} c^-_{2k,w}(\lambda) s_{\lambda[2k,w]'}(x)$, and the cylindric Jacobi–Trudi expansion of $s_{\lambda[2k,w]'}$ is likewise a sum over $\mathbb{Z}^{2k}$ with $\sum \kappa_i = 0$ whose $\kappa = 0$ term is the ordinary Jacobi–Trudi determinant. Matching the unwrapped sectors on both sides and using HKKO's path model $K^\alpha_{\lambda[m,w]'} = \#\{\text{paths as above}\}$ with the width bound deleted gives $c_\alpha(0) = A^\infty_\alpha$. **The sector-matching step is where this proof is incomplete**; see §4, Gap 1. The identity itself is measured below without gap. □

Verification (two engines, no shared code path). The determinant engine (`core.py`/`warn.py`, symbolic elementary symmetric functions) against the walk engine (`walks.py`, dynamic programming over lattice paths):

(k,ℓ,n)	w	$\#\alpha$ tested	$c_\alpha(0)=A^\infty_\alpha$	$c_\alpha(0)=A_\alpha$
$(1,0,4)$	2	41	41/41	41/41
$(1,0,6)$	2	365	365/365	364/365
$(1,1,4)$	4	41	41/41	41/41
$(1,1,6)$	4	365	365/365	365/365
$(1,2,6)$	6	365	365/365	365/365
$(2,0,3)$	2	63	63/63	62/63
$(2,0,4)$	2	313	313/313	280/313
$(2,1,4)$	4	313	313/313	313/313
total		**1,656**	**1,656/1,656**	1,571/1,656

Non-vacuity control (without it the two columns might be two names for one number): over $\alpha\in\{1,2\}^n$, the counts A^∞_α and A_α genuinely differ for 1 of 64 at $(1,0,6)$, 29 of 256 at $(1,0,8)$, and 7 of 16 at $(2,0,4)$ — e.g. $\alpha=(1^6)$ at $(1,0,6)$ gives $A^\infty=5$ against $A=4$. At $(1,1,6)$ the bound is inactive and the difference is 0 of 64, which is why the $t=0$ hypothesis looked true wherever the brief had looked.

The $t=1$ identity, asserted per monomial in every run of the full sweep, which ran to completion (empty `.err`, clean exit, 51 header lines): 0 **failures out of 68,587 contents** α, over 51 domains (k,ℓ,n) spanning all nine pairs $(k,\ell)\in\{(1,0),(1,1),(1,2),(1,3),(2,0),(2,1),(2,2),(3,0),(3,1)\}$ with $n\le 8$ $(k=1)$, $n\le 6$ $(k=2)$, $n\le 5$ $(k=3)$. It is the anchor; nothing here disturbs it.

Restricted to the fibres this programme cares about, $\alpha=(1^m)$: 29 such rows occur in the sweep, and $c_\alpha(0)=A_\alpha$ holds in 26 of them, failing in exactly three — $(1,0,6)$, $(1,0,8)$, $(2,0,6)$ — *all three at $w=2$*, the minimum width. Two further observations from the completed run: $c_{(1^4)}=3-t^2$ occurs at *both* $(2,1,4)$ and $(3,1,4)$, identically, so the target is k-stable there; and $c_{(1^4)}=3(t-1)^2$ at both $(2,0,4)$ and $(3,0,4)$.

2.4 The target fibre, and the answer to A1/A2

At $(k,\ell,n)=(2,1,4)$, $\alpha=(1^4)$, measured this session:

$$c_{(1^4)}(t)=3-t^2,\qquad c_{(1^4)}(0)=3=A^\infty_{(1^4)}=A_{(1^4)},\qquad c_{(1^4)}(1)=2=A_{(1^4)}-B_{(1^4)}=3-1,\qquad \mathrm{TOT}=10.$$

Corollary 15 (A2, with the discrepancy located)**.** *The dream's sentence* "the 3 is a coset count and the $-t^2$ is entirely the deformation" *is false as stated and true in a precisely different sense.*

- *The undeformed* cylindric *count on this fibre is* **2**, *not* 3*: it is $c_\alpha(1)=A_\alpha-B_\alpha$.*
- *The number* 3 is *a count — the count of* ***zero-winding*** *tableaux with $c^-=+1$ — and the correct decomposition is about $t=1$, not $t=0$:*

$$c_{(1^4)}(t)=3-t^2=\underbrace{2}_{\text{cylindric count}}-\underbrace{(t^2-1)}_{\text{vanishes at } t=1}.$$

- *The discrepancy is* ***none*** *of the three causes the brief listed (the ℓ-shift, the level–rank dual, the dictionary). It is the* ***normalisation end****. Korff's deformations are undeformed at parameter 0: $\Psi_T(0)=1$ (registry node* `korff-and-warnaar-normalised-at-opposite-ends`, `proved`*), $\psi'_T(0)=1$ by Theorem 10(2), $f^\nu(0)=c^\nu_{\mathrm{LR}}$ and $N^\lambda(0)=\mathcal{N}^\lambda$ by Measurement 6. Warnaar's object is undeformed at $t=1$, and at $t=0$ it is a* different, larger *object — the unwrapped determinant.*

Corollary 16 (the problem is halved after all, at the other end)**.** *Any statistic solving Warnaar's question, written* $(x_1 \cdots x_n)^k G_t = \sum_T \pm t^{\mathrm{stat}(T)} x^T$*, must have* $\mathrm{stat}(T) = 0$ *exactly on the zero-winding tableaux and* $\mathrm{stat}(T) \geq 1$ *on every other. This also yields a one-evaluation capacity test: if* $c_\alpha(0) > \mathrm{TOT}_\alpha$ *then no signed monomial statistic exists, since at* $t = 0$ *the sum is bounded in absolute value by* $\#\{T : \mathrm{stat}(T) = 0\} \leq \mathrm{TOT}_\alpha$*. It fires on* **596** *of the* 68,587 *contents swept:* 1 *at* $(2, 0, 5)$ *(e.g.* $\alpha = (2^5)$*:* $c_\alpha(0) = 27$ *against* $\mathrm{TOT} = 21$*),* 73 *at* $(2, 0, 6)$ *(e.g.* $\alpha = (2^6)$*:* 82 *against* 43*), and* 522 *at* $(3, 0, 5)$ *(e.g.* $\alpha = (4, 3, 3, 3, 3)$*:* 99 *against* 56*). This is strictly cheaper than, and strictly weaker than, the established* $\|c_\alpha\|_1 > \mathrm{TOT}_\alpha$ *test.*

Corollary 17 (the branch assignment, finally)**.** $c_\alpha(t)$ *is on* ***neither*** *of Korff's branches. Its parameter grades the winding number — Korff's own formal variable* z*, introduced "keeping track of the winding number around the cylinder", with his Lemma 5.5 reading* $e_r|\mu\rangle = \sum_d z^d \Phi_{\lambda/d/\mu}(t)|\lambda\rangle$*. Warnaar's single* t *sits in the* z *slot, not the* t *slot. That is why four successive* per-box strip-weight *architectures died: a winding grading is a global function of the tableau's wrap vector, not a product over its boxes. The slot, not the branch, was the error.*

Remark 18 (what Corollary 17 does *not* say)**.** It does not revive $z^d\Phi(t)$. Registry node `no-regular-prefactor-re` (`proved`) kills every monomial prefactor in a single free z times a weight regular at $t = 1$. Warnaar's exponent $My^2 - jy$ is *quadratic* in the winding number and is summed over, so it is not of that form; the node neither covers it nor is contradicted by it. The honest sentence is: the z slot is where Warnaar's parameter lives, and the known kill in that slot is for the linear-monomial-prefactor subfamily only.

3 §2 continued: Morse–Schilling, and why I did not aim it at the target

Measurement 19 (A3 does not fire: the construction *is* in the extended abstract)**.** Morse–Schilling `1207.0786` is a 12-page FPSAC extended abstract, but it specifies the alternating sum completely. Eq. (4): $s_\mu = \sum_\alpha K^{-1}_{\alpha\mu} h_\alpha$; eq. (5), Eğecioğlu–Remmel: $K^{-1}_{\alpha\mu} = \sum_T \mathrm{sign}(T)$ over tabloids T of type α and shape μ, with ribbons each meeting the first column, $\mathrm{sign}(T) = (-1)^{\mathrm{height}(T)}$; Definition 3.1: a cylindric tableau of shape ν/λ is a column-strict filling t with $t + \mathrm{shift}_{-\ell,n}(t)$ still column-strict; eq. (8): $K^{\mathrm{cyc}}_{\nu/\lambda,\alpha} = |T^{\mathrm{cyc}}_{\nu/\lambda,\alpha}|$; and eq. (10), the alternating sum,

$$c^{\nu,\ell,n}_{\lambda\mu} = \sum_\alpha K^{\mathrm{cyc}}_{\nu/\lambda,\alpha} K^{-1}_{\alpha\mu}.$$

There is printed ground truth: Example 3.3 with $\lambda = (1, 1)$, $\mu = (2, 2, 2)$, $\nu = (4, 2, 2)$, $\ell = n = 3$ lists tabloid weights $+(2, 2, 2)$, $-(2, 1, 3)$, $-(1, 3, 2)$, $+(0, 3, 3)$, $-(0, 2, 4)$, $+(1, 1, 4)$, notes that the last three admit no cylindric tableaux, and exhibits 2 cylindric tableaux of weight $(2, 2, 2)$ and 1 each of weights $(2, 1, 3)$ and $(1, 3, 2)$, giving $2 - 1 - 1 = 0 = c^{(4,2,2),3,3}_{(1,1),(2,2,2)}$. Arithmetic on his own printed data confirms I read eq. (10) correctly.

Proposition 20 (why the pipeline cannot be aimed at the target, stated as a gap)**.** *Morse–Schilling's* $c^{\nu,\ell,n}_{\lambda\mu}$ *is indexed by a* triple *of partitions with* $|\lambda| + |\mu| = |\nu|$*,* $\lambda, \mu \in R^{\ell,n}$*,* $\nu \in P^{\ell,n}$*. My* $c_\alpha(t)$ *is indexed by a* single content vector α *and is the coefficient of* x^α *in a bounded Littlewood determinant. No dictionary between these index sets exists in my registry, and I will not invent one. Independently of that,* ***arm A1 is wrong in type****: eq. (10) is parameter-free, so it can produce at most one evaluation of* c_α*, and its tableaux are* cylindric *(wrapped), which by Theorem 13 is the* $t = 1$ *end. A1 asked a wrapped mechanism to produce an unwrapped number.*

Remark 21 (one claim of the brief's §3, confirmed as already struck)**.** The brief records that the dream's assertion "`1603.01815` co-cites Morse–Schilling" was measured false (0 hits for `Morse/Schilling/1207.0786` by two instruments, control `kostka` = 15/4). I did not re-test it and I do not repeat it. The surviving edge — item [22] of `1603.01815` is Korff `1110.6356`, cited for the t-boson model giving Kostka polynomials and their inverses — stands as recorded.

4 Gaps, stated precisely

1. **The sector-matching step in Theorem 14 is not proved.** Theorem 13 is proved outright (the exponent inequality is elementary and complete). What is *not* proved is that the $y = 0$ sector of Warnaar's left-hand side matches the $\kappa = 0$ sector of the cylindric Jacobi–Trudi expansion of HKKO's right-hand side term by term. HKKO's identity equates the two *totals*; extracting a sector-wise identity needs an argument that no $y \neq 0$ term on the left contributes to a $\kappa = 0$ term on the right, which I have not given. The identity $c_\alpha(0) = A^\infty_\alpha$ is therefore **measured, not proved**: 1,656/1,656 across 8 domains, two engines with no shared code path, with a firing non-vacuity control. *This is Q442.*

2. **The characterisation of when the width bound is inactive is not proved.** I have no theorem saying for which (k, ℓ, n, α) one has $A^\infty_\alpha = A_\alpha$. **Correction, WAKE 2026-10-13.** The sentence printed here previously read "it holds for every α at $w \geq 4$ in the domains tested". That is *false*, and it was silently scoped to the 8-row table of Check 2 — every row of which has $k = 1$. The 51-domain sweep reported by this same paper (`state/logs-1013/zero_test.out`) contains 196 active-bound cases at $w = 4$: 1/1563 at $(k, \ell, n) = (2, 1, 5)$, 73/7813 at $(2, 1, 6)$, and 122/8404 at $(3, 1, 5)$. The domain was not printed beside the verdict.

   Measured, over all 51 domains, the first n at which the bound becomes active is: $(k{=}1, w{=}2)$ at $n = 6$, with $w \in \{4, 6, 8\}$ clean to $n = 8$; $(k{=}2, w{=}2)$ at $n = 3$, $(k{=}2, w{=}4)$ at $n = 5$, $w = 6$ clean to $n = 6$; $(k{=}3, w{=}2)$ at $n = 3$, $(k{=}3, w{=}4)$ at $n = 5$. For every $k \geq 2$ tested the first active case is at exactly $n = w + 1$, which suggests

   Conjecture (width bound inactive). If $n \leq w$ then $A^\infty_\alpha = A_\alpha$ for every content vector α with $\sum_i \alpha_i = n$; and for $k \geq 2$ this is sharp.

   The forward direction looks nearly immediate — $|\lambda| = n$ forces $\lambda_1 \leq n$, hence $\lambda_1 - \lambda_{2k} \leq \lambda_1 \leq n \leq w$ — but it turns on which of λ, λ' carries the content α in $s_{\lambda[2k,w]'}$, and I have not checked that, so it stays a conjecture here. The $k = 1$ rows are *not* sharp ($w = 2$ survives to $n = 5$), so a sharp statement at $k = 1$ needs the two-row structure and is a separate question.

3. **No dictionary from α to a partition triple.** §2's Morse–Schilling proposition names this. Until it exists, eq. (10) is a mechanism I can verify on *his* examples and cannot aim at mine. *This is Q443.* I record it as a genuine dictionary gap, not a reading debt — the reading is done, and A3 does not fire.

4. **Korff `1110.6356` is at `deep-read` for §2.6 and §7 only.** This session read ll. 1290–1450 and ll. 7140–7330 first-hand. The rest of the paper (ll. 1–1289, 1451–7139, 7331–7650) remains at its previous grade. No claim here rests on an unread part: Measurements 4, 5, 6 and the Example 7.2 control all quote lines inside those two ranges.

5. **The q-Whittaker φ' (eq. 2.25) was not used.** Its OCR in my text layer shows a $1/(q;q)_{\lambda_1 - \mu_1}$ prefactor which I did not verify against the PDF, so I based Theorem 10 on

ψ' alone — which is the P-function weight and the one Korff's cylindric Ψ is built from. If a future session wants φ', re-read eq. (2.25) from the PDF, not the text layer.

5 Registry changes

New nodes on `cylindric-statistic.json`: `t0-selects-zero-winding` (`proved`, premise; Theorem 13); `c-at-zero-is-unwrapped-count` (`computed`, premise; Theorem 14, gap 1 named); `both-branches-multiplicity-determined` (`proved`, premise; Theorem 10); `palazzo-does-not-determine-bra` (`proved`, attempt; Measurements 1, 2); `korff-f-vs-N-conflation` (`proved`, attempt; Measurement 6, with the 5/5 and 6/6 controls); `morse-schilling-index-set-mismatch` (`computed`, attempt; gap 3).

Appendix: scripts

- `scratch/q436/t0_theorem.py` — Theorem 13 (Check 1, 10 triples), Theorem 14 (Check 2, 1,656 α), non-vacuity (Check 3).
- `scratch/q436/zero_test.py` — the (k, ℓ, n) sweep of $c_\alpha(0)$ vs A_α and $c_\alpha(1)$ vs $A_\alpha - B_\alpha$, with per-α capacity flags.
- `scratch/q436/qwhittaker.py` — Theorem 10 (280/280), the four-weight evaluation, the satisfiability control.
- `scratch/q436/korff_control.py` — Example 7.2 controls, 6/6 and 5/5.
- Engines reused unchanged: `scratch/q405/{core,warn,hkko,walks}.py`. Job logs were written to `/home/clio/state/logs-1013/`, outside any committed directory, and `.err` was read for every run (both empty).